

EV RANGE PREDICTION USING MACHINE LEARNING – PROJECT REPORT

1. Introduction

This project predicts the driving range of Electric Vehicles (EVs) using a Kaggle dataset.

Machine Learning models are applied to understand how factors like speed, temperature, battery level, and driving behaviour affect the EV range.

2. Objectives

- To analyze EV driving data.
- To preprocess and clean the dataset.
- To train ML models to predict EV range.
- To evaluate model accuracy and identify key features.

3. Tools and Technologies

- Google Colab
- Python
- Pandas, NumPy
- Scikit-Learn
- Matplotlib, Seaborn
- GitHub for project hosting

4. Methodology

- Dataset loading and cleaning.
- Exploratory Data Analysis (EDA).
- Feature engineering.
- Model training (Linear Regression, Random Forest, XGBoost).
- Performance evaluation using RMSE and accuracy metrics.

5. Results

Tree-based models like Random Forest and XGBoost produced high accuracy.

The most important features influencing EV range were speed, temperature, and battery percentage.

6. Conclusion

This project demonstrates that machine learning can effectively predict EV range and help reduce range anxiety. It strengthens practical ML skills and is suitable for industry-level applications in electric mobility.